

Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

§6.1. Digits of a number

(6.1a)

$\text{val}^\circ = \text{embed}^\circ \text{wrap} \cup \text{op}^\circ - \times \text{Digit}(\text{val}^\circ) \text{snoc}$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(a)	val°	the converse of val , which reads digits as a number
(b)	$([\text{wrap}, \text{snoc}]^\circ \text{Digit}^{++} (-\times \text{Digit}(\text{val})) [\text{embed}, \text{op}])^\circ$	val undoes wrap or snoc , reads the front with val and combines by embed or op , so its converse runs that backwards
(c)	$(\text{wrap}^\circ \text{embed})^\circ \cup (\text{snoc}^\circ - \times \text{Digit}(\text{val}) \text{op})^\circ$	undoing wrap or snoc first gives two cases: a decimal and its last digit read by val then op , or one digit read by embed
(d)	$\text{embed}^\circ \text{wrap} \cup \text{op}^\circ - \times \text{Digit}(\text{val})^\circ \text{snoc}$	the converse of each case is its steps undone in reverse: undo op , undo reading the front, then snoc ; or undo embed , then wrap
(e)	$\text{embed}^\circ \text{wrap} \cup \text{op}^\circ - \times \text{Digit}(\text{val}^\circ) \text{snoc}$	undoing the read of the front while keeping the last digit is val° on the front alone

(6.1b)

$$\text{op}^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$$

op° splits a number into its quotient and remainder by 10

$$(\exists p. 0 < \pi_1 p \wedge \text{op}^\circ(m, p)) \iff 10 \leq m$$

op° gives a pair with a nonzero first component exactly at the numbers with two or more digits

$$(\exists d. \text{embed}^\circ(m, d)) \iff 0 < m \wedge m < 10$$

embed° gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

(6.2b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
$\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation

$$\alpha^\circ F(X) R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$$

(6.2c)

a relation X that contains one unfolding of the fold's recursion at X contains the fold

	(a)	$\langle R \rangle$	
$=$	(b)	$(\mu X : \alpha^\circ F(X) R)$	$\langle R \rangle$ the least fixed point — Theorem 6.1, (6.2b)
$\sqsubseteq$	(c)	X	$\alpha^\circ F(X) R \sqsubseteq X$, μ below every prefix point

$$X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq \langle R \rangle$$

(6.2d)

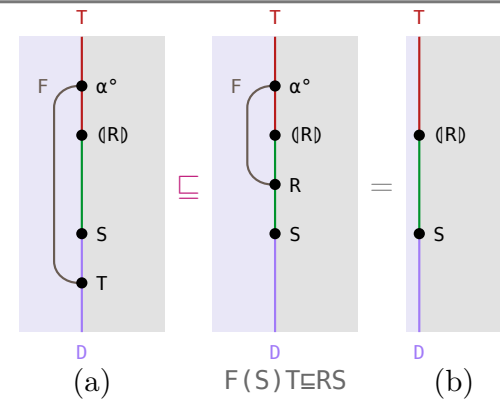
a relation X contained in one unfolding of the fold's recursion at X is contained in the fold

	(a)	X	
$\sqsubseteq$	(b)	$(\nu X : \alpha^\circ F(X) R)$	$X \sqsubseteq \alpha^\circ F(X) R$, ν above every postfix point
$=$	(c)	$\langle R \rangle$	$\langle R \rangle$ the greatest fixed point

$$F(S) T \sqsubseteq R \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$$

(6.2e)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S

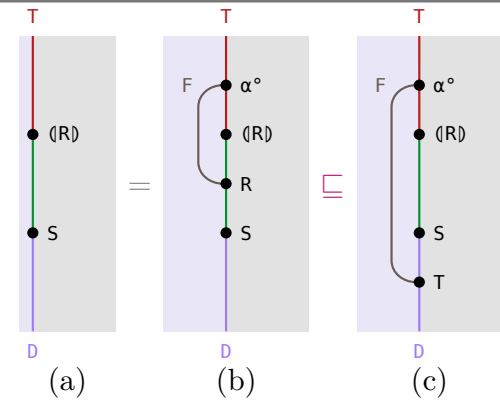


$F(RS) = F(R) F(S)$ — $\triangleq$ relator

$$R \sqsubseteq F(S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$$

(6.2f)

if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T



§6.3. Hylomorphisms

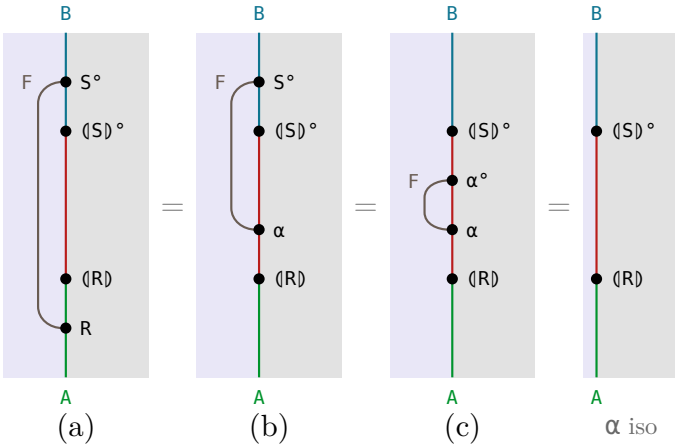
$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : \mathbb{S}^\circ F(X)R)$$

(6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

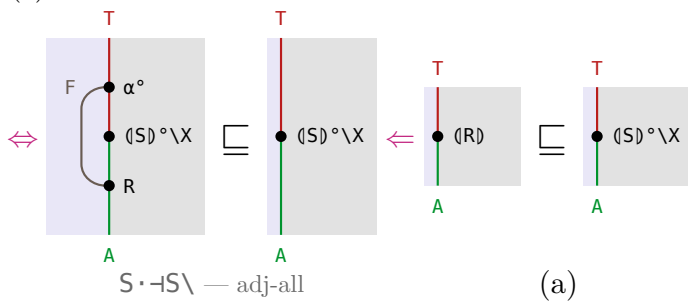
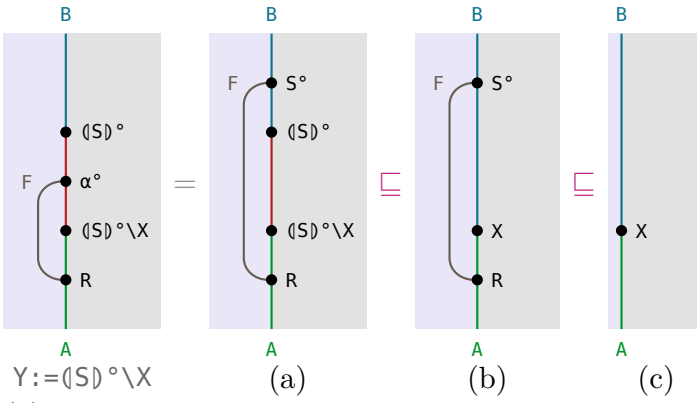
$$\mathbb{S}^\circ F((\mathbb{S})^\circ(\mathbb{R}))R = (\mathbb{S})^\circ(\mathbb{R})$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term $\mathbb{S}^\circ$ represents the decomposition stage, $F((\mathbb{S})^\circ(\mathbb{R}))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $\mathbb{S} : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial



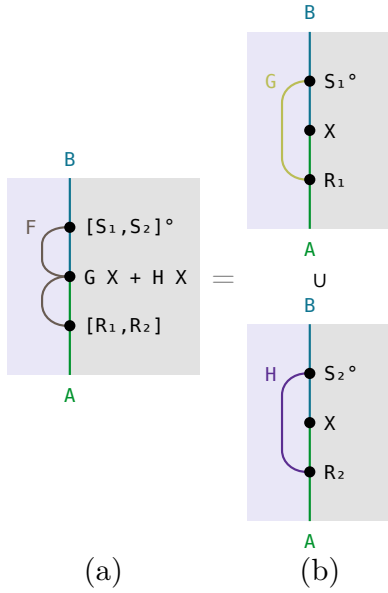
$$\mathbb{S}^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ(\mathbb{R})$ is included in X if X satisfies the associated recursion inequation



$$(\forall x. \forall y. \forall X. F(X)=G(X)+H(X)) \Rightarrow ([S_1, S_2]^\circ \circ ([R_1, R_2])^\circ) = (\mu X : S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

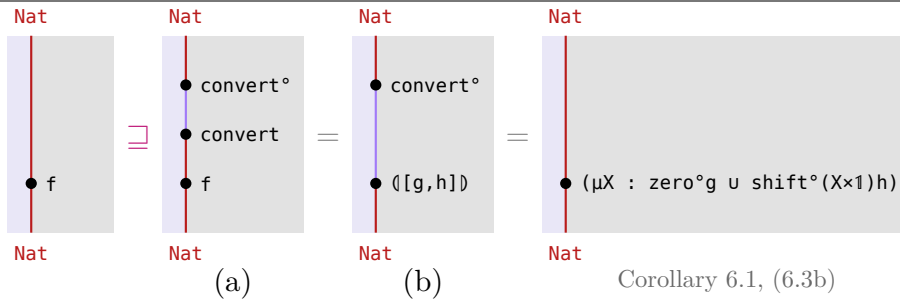


(a)	$[S_1, S_2]^\circ (G(X)+H(X)) [R_1, R_2]$	the body of Theorem 6.2 at $F(X)=G(X)+H(X)$
(b)	$S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2$	$[S, T] = (S+T) \nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$; the $+$ stands for the codiagonals ∇°, ∇ , which are not drawn $\nabla^\circ (P+Q) \nabla = P \cup Q$ — coproduct

§6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h \Rightarrow (\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1) h) \sqsubseteq f \quad (6.4a)$$

once **f** sends **zero** to **g** and turns **shift** into **h**, the recursion that halves the argument at each step computes **f**



Corollary 6.1, (6.3b)

- (a) convert simple
(b) fusion: $\text{zero } f = g, \text{shift } f = (f \times 1) h$

$$(\mu X : \text{zero}^\circ \text{one} \cup \text{shift}^\circ (X \times 1) \text{op}(a)) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes **a** to the power **b**

$\text{zero } \text{exp}(a) = \text{one}$

a to the power **0** is **1**

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

a to the power **2n+d** is **op(a)** of **a** to the power **n** and **d**

$$(\mu X : \text{zero}^\circ \text{zero} \cup \text{shift}^\circ (X \times 1) \text{op}(b)) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves **a** at each step computes the remainder of **a** by **b**

$\text{zero } \text{mod}(b) = \text{zero}$

0 mod b is **0**

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

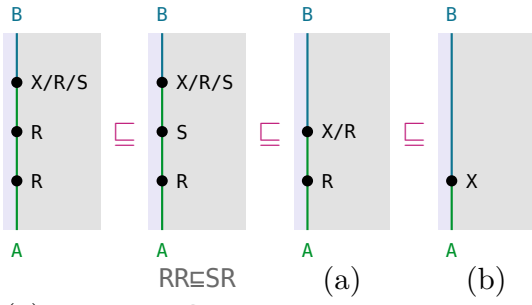
(2a+d) mod b is **op(b)** of **a mod b** and **d**

§6.5. Unique fixed points

$$\text{inductive}(S) \wedge R \sqsubseteq SR \Rightarrow \text{inductive}(R)$$

(6.5a)

R is inductive when $R \sqsubseteq SR$ for an inductive S



- (a) division by S
(b) division by R

$$\text{member} = \emptyset$$

(6.5b)

a constant relator records no elements, so its membership is empty

	(a)	member	
=	(b)	$\emptyset$	

$$\text{member} = [\text{member}, \text{member}]$$

a member of $F+G$ is a member of whichever summand the value is in

	(a)	member	
=	(b)	$[\text{member}, \text{member}]$	

$$\text{member} = \pi_1 \text{ member} \cup \pi_2 \text{ member}$$

a member of $F \times G$ is a member of either component

	(a)	member	
=	(b)	$\pi_1 \text{ member} \cup \pi_2 \text{ member}$	

$$\text{inductive}(\text{Smember}) \wedge X = \text{SF}(X)R \wedge Y = \text{SF}(Y)R \Rightarrow X = Y$$

(6.5c)

when S **member** is inductive, two solutions X, Y of $X = \text{SF}(X)R$ are equal

$$\text{inductive}(\text{Smember}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge \text{SF}(X)R \sqsubseteq X \Rightarrow \text{Entire}(X)$$

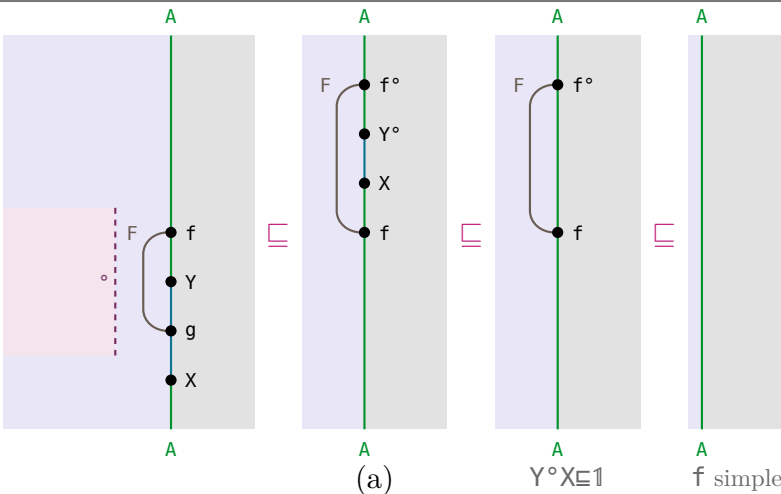
(6.5d)

when S **member** is inductive and S, R are entire, a pre-fixed point of $X \mapsto \text{SF}(X)R$ is entire

$$\text{inductive}(\text{gmember}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = gF(X)f \Rightarrow \text{Map}(X)$$

(6.5e)

when g **member** is inductive and f, g are maps, the solution of $X = gF(X)f$ is a map

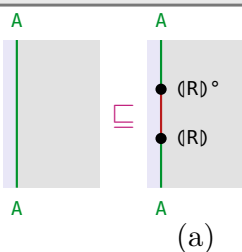


- (a) $X = gF(X)f$, g simple

$$\text{inductive}(R^\circ \text{member}) \wedge 1 \sqsubseteq R^\circ R \Rightarrow 1 \sqsubseteq (R)^\circ (R)$$

(6.5f)

when $R^\circ \text{member}$ is inductive and R is surjective, the fold (R) is surjective

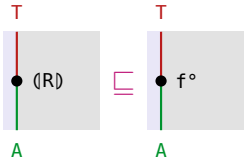


- (a) (6.5d) at $S = R^\circ$

$$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \wedge 1 \sqsubseteq R^\circ R \Rightarrow f^\circ = \langle R \rangle$$

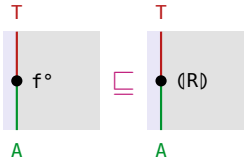
(6.5g)

a map f with $Rf \sqsubseteq F(f)\alpha$, for a surjective R , has the fold $\langle R \rangle$ as its converse



(a)

(a) shunting f , fusion, $Rf \sqsubseteq F(f)\alpha$



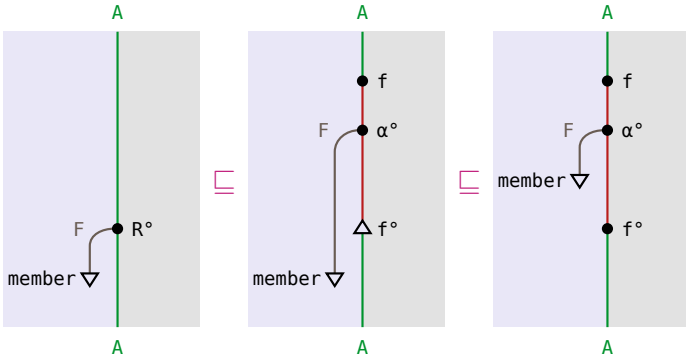
(a)

(a) (6.5f) and $\langle R \rangle f \sqsubseteq 1$

$$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \Rightarrow R^\circ \text{member} \sqsubseteq f\alpha^\circ \text{member } f^\circ$$

(6.5h)

$R^\circ \text{member}$ is below $\alpha^\circ \text{member}$ conjugated by f , so it is inductive when $\alpha^\circ \text{member}$ is



(a)

(b)

(a) $Rf \sqsubseteq F(f)\alpha$, shunting

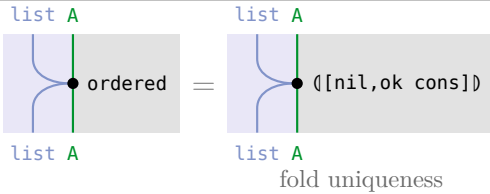
(b) member lax natural

§6.6. Sorting by selection

$$\text{ordered} = \langle [\text{nil}, \text{ok cons}] \rangle$$

(6.6a)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

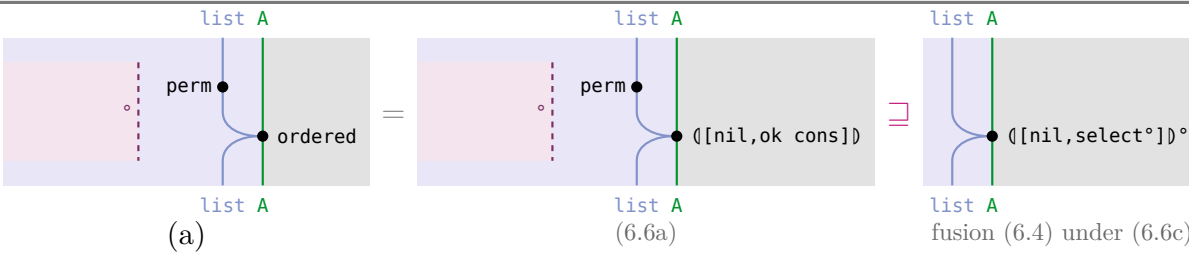


fold uniqueness

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow \langle [\text{nil}, \text{select}^\circ] \rangle^\circ \sqsubseteq \text{perm ordered}$$

(6.6b)

every output of unfolding the input by **select** is a sorted permutation of it



(a)

(6.6a)

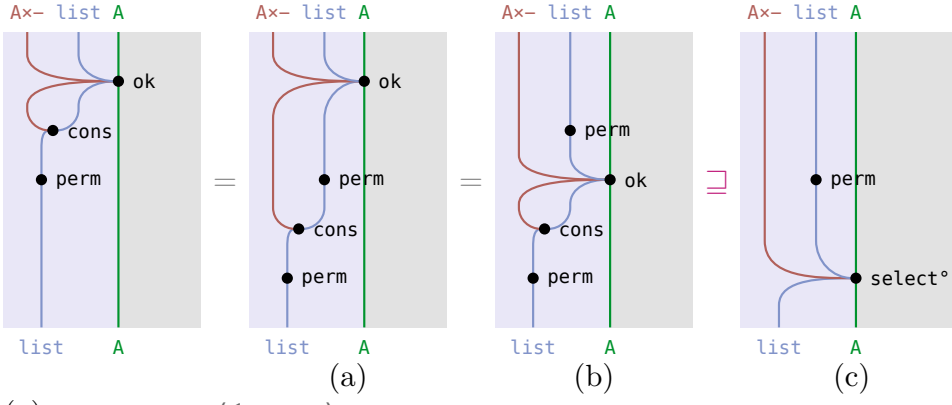
fusion (6.4) under (6.6c)

(a)

$\text{perm}^\circ = \text{perm}$, $\text{ordered}^\circ = \text{ordered}$

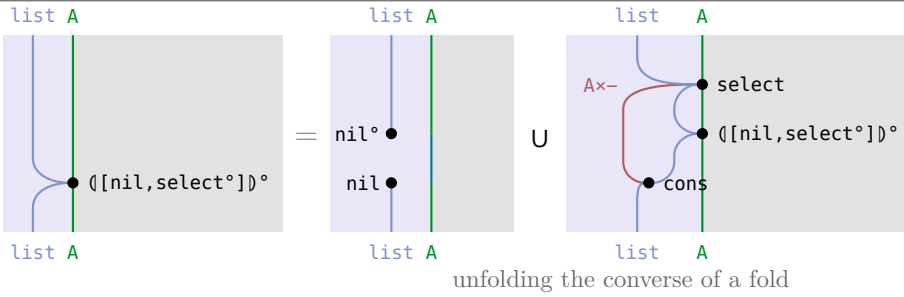
$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6c)$$

permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



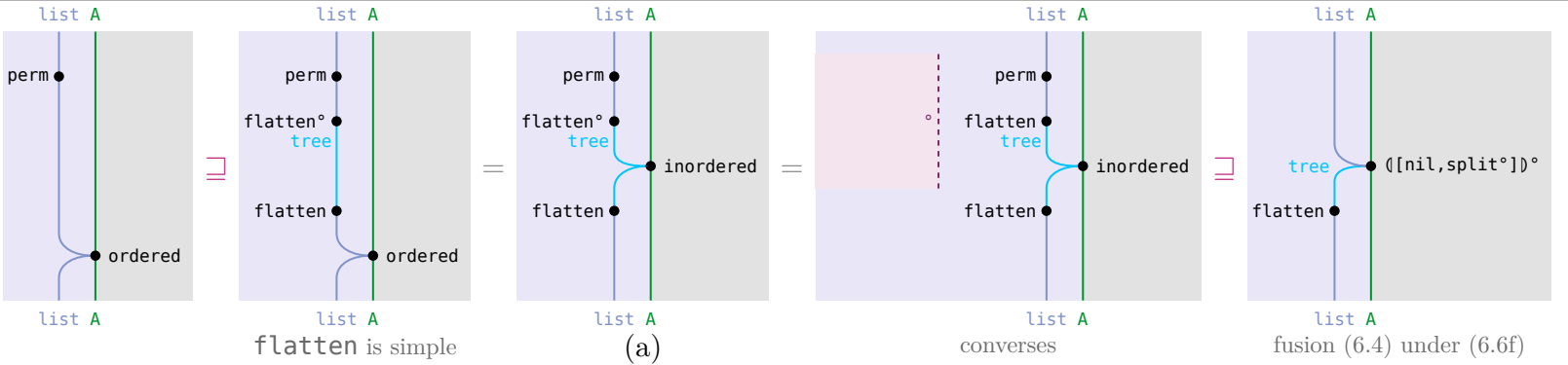
$$([nil, \text{select}^\circ])^\circ = nil^\circ nil \vee \text{select} (1 \times ([nil, \text{select}^\circ])^\circ) \text{cons} \quad (6.6d)$$

selection sort returns $[]$ on $[]$, and otherwise selects (a, y) , sorts y and conses a back on



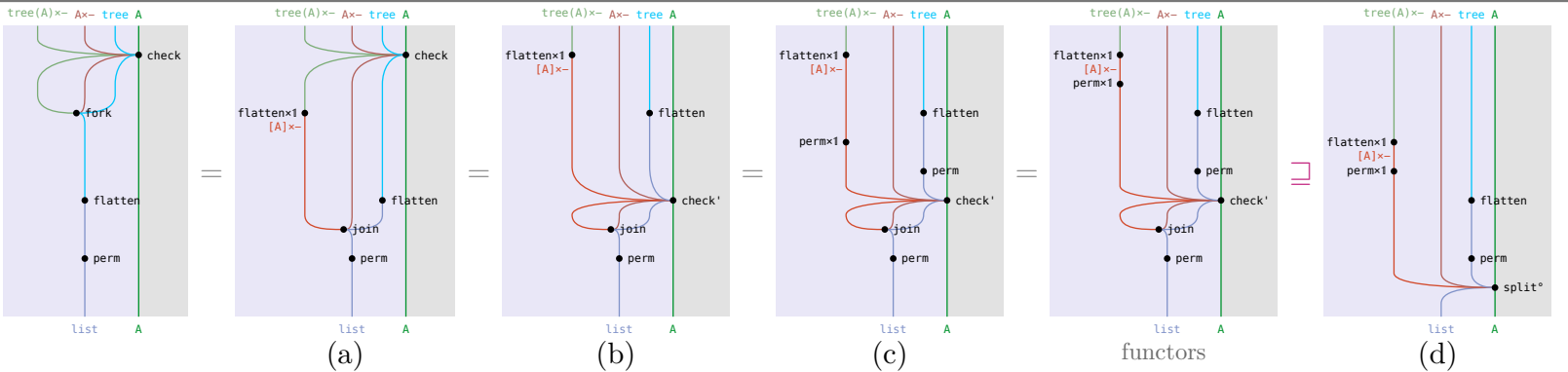
$$(\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ([nil, \text{split}^\circ])^\circ \text{flatten} \sqsubseteq \text{perm ordered} \quad (6.6e)$$

every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ \sqsubseteq \text{check}' \text{ fork flatten perm} \quad (6.6f)$$

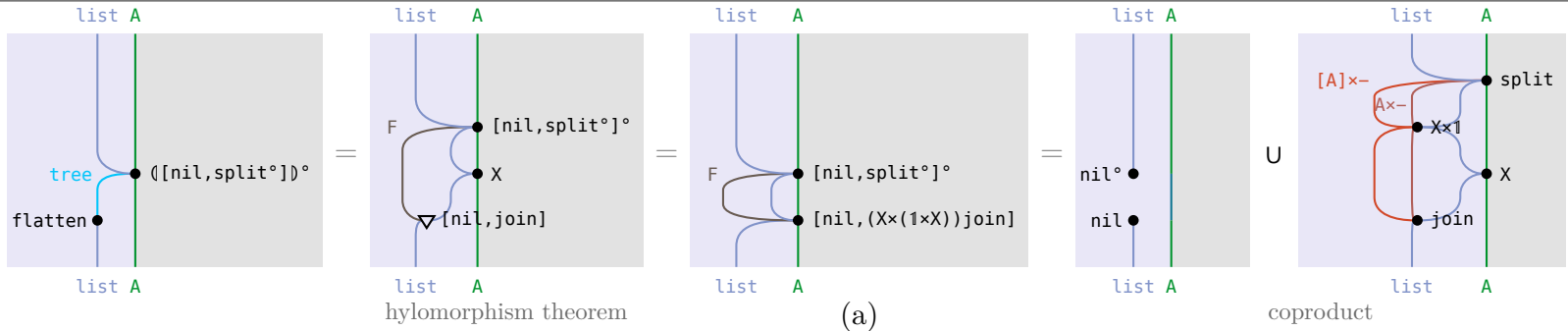
undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted



$$([nil, split^\circ])^\circ \circ flatten = nil^\circ nil \cup split \quad (6.6g)$$

$$((([nil, split^\circ])^\circ \circ flatten) \times (1 \times ([nil, split^\circ])^\circ \circ flatten))) \circ join$$

quicksort returns [] on [], and otherwise splits into (x, a, y), sorts x and y and joins them around a



§6.7. Closure

$$R^* = (\mu X : 1 \cup XR) \quad (6.7a)$$

closing R by composing it on the left and closing it by composing on the right give the same relation R^*

	(a)	R^*	
=	(b)	$(\mu X : 1 \cup XR)$	(6.8), Ex 6.31

$$1 \in X \wedge XX \in X \implies R \in X \iff R^* \in X \quad (6.7b)$$

for a preorder X , X contains R exactly when it contains R^*

$$1 \in R^* \quad (6.7c)$$

R^* is reflexive

	(a)	1	
$\sqsubseteq$	(b)	$1 \cup RR^*$	union
=	(c)	R^*	fixed point, Thm 6.1

$$R \in R^* \quad (6.7d)$$

R^* contains R

	(a)	R	
=	(b)	R	identity
$\sqsubseteq$	(c)	RR^*	$1 \in R^* \implies (6.7c)$
$\sqsubseteq$	(d)	R^*	fixed point

$$R^* R^* \in R^* \quad (6.7e)$$

R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto 1 \cup RX$

	(a)	$(1 \cup R(R^*/R^*))R^*$	
=	(b)	$R^* \cup R(R^*/R^*)R^*$	$(P \cup Q)T = PT \cup QT$ — composition distributes over $\cup$
$\sqsubseteq$	(c)	$R^* \cup RR^*$	$(S/S)S \subseteq S$
$\sqsubseteq$	(d)	R^*	$RR^* \in R^* \implies (6.7d)$

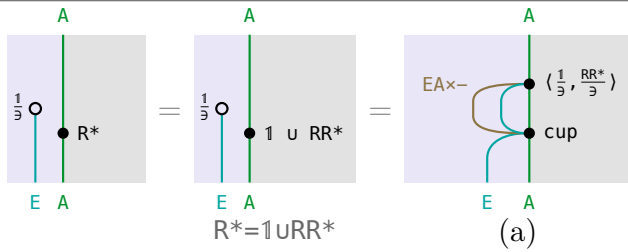
$$1 \in X \wedge XX \in X \wedge R \in X \implies R^* \in X \quad (6.7f)$$

a reflexive transitive X containing R is a prefixed point of $X \mapsto 1 \cup RX$, so it contains R^*

	(a)	$1 \cup RX$	
$\sqsubseteq$	(b)	$X \cup XX$	$1 \in X, R \in X$
$\sqsubseteq$	(c)	X	$XX \in X$

$$\frac{R^*}{S} = \left(\frac{1}{S}, \frac{RR^*}{S} \right) \text{cup} \quad (6.7g)$$

the set of R^* -successors of a is a itself joined with the R^* -successors of its R -successors



(a) $\wedge(R \cup S) = \langle \wedge(R), \wedge(S) \rangle \text{cup}$

$$R - \emptyset = R \quad (6.7h)$$

taking nothing away leaves R

$$R \cup (S - R) = R \cup S$$

beside R , only the part of S outside R adds anything

$$R - (S \cup T) = R - S - T$$

taking away S and then T takes away $S \cup T$

$$(R \cup S) - T = (R - T) \cup (S - T)$$

taking T away from a union takes it away from each part

$$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y))) \quad (6.7i)$$

the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ

	(a)	$(\mu X : \varphi(\psi(X)))$	
=	(b)	$\varphi(\mu Y : \psi(\varphi(Y)))$	Ex 6.35

$$SR^* = (\mu X : S \cup XR) \quad (6.7j)$$

S followed by any number of R steps is the least X containing S and closed under a further R step

	(a)	SR^*	
=	(b)	$(\mu X : S \cup XR)$	p.160

$$R^*S = (\mu X : S \cup RX)$$

any number of R steps followed by S is the least X containing S and closed under an R step in front

	(a)	R^*S	
=	(b)	$(\mu X : S \cup RX)$	Ex 6.32

$$\theta(\emptyset, S) = SR^* \quad (6.7k)$$

started with nothing found and S to explore, θ computes SR^*

	(a)	$\theta(\emptyset, S)$	
=	(b)	$\emptyset \cup (\mu X : S \cup (XR - \emptyset))$	definition of θ
=	(c)	$(\mu X : S \cup XR)$	$R - \emptyset = R$ — (6.7h)
=	(d)	SR^*	(6.7j)

$$\theta(P, \emptyset) = P \quad (6.7l)$$

with nothing left to explore, θ returns what it has found

	(a)	$\theta(P, \emptyset)$	
=	(b)	$P \cup (\mu X : \emptyset \cup (XR - P))$	definition of θ
=	(c)	$P \cup \emptyset$	$\emptyset$ is a prefixed point: $\emptyset R - P = \emptyset$
=	(d)	P	$P \cup \emptyset = P$